

1. What is the difference between global and local scope of a variable?

Answer:

The **scope of a variable** means the part of the program where the variable can be accessed.

Global Variable

Defined outside a function.

Can be accessed throughout the program.

Exists for the entire program execution.

Local Variable

Defined inside a function.

Can be accessed only inside that function.

Exists only while the function is executing.

Example:

```
x = 10    # Global variable
```

```
def show():
```

```
    y = 20 # Local variable
```

```
    print(x)
```

```
    print(y)
```

```
show()
```

Here, x is a **global variable** and y is a **local variable**.

2. What is a recursive function?

Answer:

A **recursive function** is a function that **calls itself** to solve a problem. It must have a **base condition** to stop the recursion.

Example:

```
def factorial(n):
```

```
    if n == 1:
```

```
        return 1
```

```
    return n * factorial(n - 1)
```

```
print(factorial(5))
```

Output:

120

Here, factorial() calls itself until the condition `n == 1` is reached.